NDID Screen Flow — Sawad Loan Universal

As of 27 Oct 2026 · uat ver222

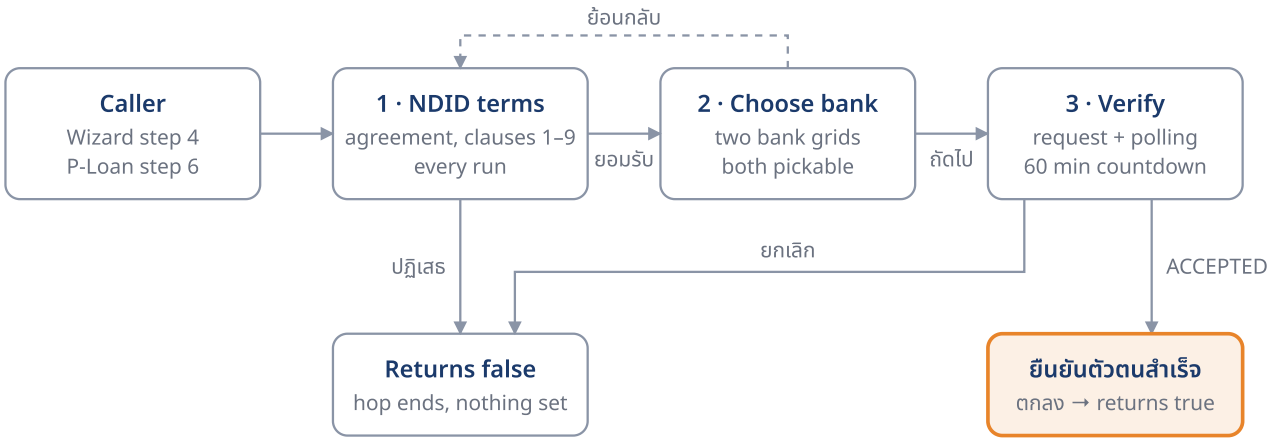
Overview

NDID is a three-screen hop: **Terms → choose a bank → verify**. It returns a single yes/no (`true` / `false`) to the screen that opened it. The customer accepts the NDID agreement, picks the bank (IdP) they authenticate with, and approves the request in that bank's own app within 60 minutes. The bank app's screens belong to the bank, so this app does not build them.

#	Screen	File	Route
1	เงื่อนไขและข้อตกลงที่เกี่ยวข้อง NDID	`ndid_terms_page.dart`	`/ndidTermsPage`
2	เลือกผู้ให้บริการยืนยันตัวตน NDID	`ndid_bank_select_page.dart`	`/ndidBankSelectPage`
3	ยืนยันตัวตน → ยืนยันตัวตนสำเร็จ	`ndid_verify_page.dart`	`/ndidVerifyPage`

All three screens are under `lib/loan_register/` . Screens 2 and 3 take an `NdidSubject` (Thai ID + chosen IdP), so the wizard and P-Loan share them.

Three screens, one bool back to the caller



Failed or timed out: the verify screen shows the standard message and ลองใหม่ creates a new request.

NDID hop · 3 screens, 2 exits

Each screen passes the `true` from the screen after it straight back. Any other exit returns `false` , and the caller stays as it was.

Entry points

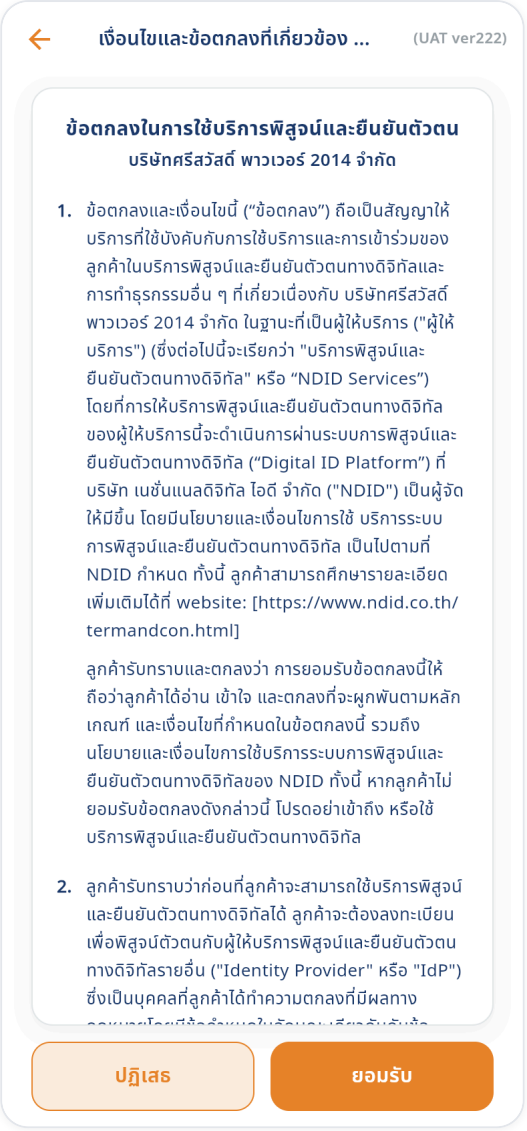
Two screens open the hop, and both push `AppRoutes.ndidTerms` and wait for one bool.

Caller	Where	Gate before opening	On `true`
Loan-register wizard	Step 4 เอกสารแนบ → ตรวจสอบเอกสาร (`document_review_page`) → **ลงนามเอกสารและยืนยันตัวตน NDID**	The customer ticks the acknowledgement checkbox on the review screen	`form.ndidVerified = true` ; step 4 shows a green check and its ถัดไป button unlocks
P-Loan	Step 6 สรุปรายละเอียดของสัญญา (step 4 in the P-Loan Extra numbering), ลงนามเอกสารและยืนยันตัวตน NDID row	The customer has accepted all 3 contract PDFs	`flow.ndidVerified = true` + `ndidReferenceId` ; this gates `canSubmit`

The top-up flow has **no** NDID step.

Screen 1 — NDID terms

The customer must accept the NDID agreement **every time** the hop runs. Acceptance is never stored, because consent is given per verification.



Screen 1 — NDID terms

- **Content:** the NDID minimum-required T&C, clauses 1–9. The text is taken word for word from the supplied document (`ndid_terms_content.dart`) and shown as one continuous scroll.
- **ยอมรับ** → logs `ndid terms accepted` → pushes Screen 2 and passes its `true` back to the caller.
- **ปฏิเสธ** → logs `ndid terms declined` → returns `false` and ends the hop.
- The decline is recorded only in this session's diagnostics trail; no server log exists yet.

Screen 2 — Choose a bank (IdP)

The customer picks one bank from two grids, and either grid can be picked from.



Screen 2 — choose a bank (plain-browser mock grid)

This is a browser capture, so the grids show the hardcoded mock banks. Inside the app they show the gateway's live list, with logos on up to 4 registered tiles.

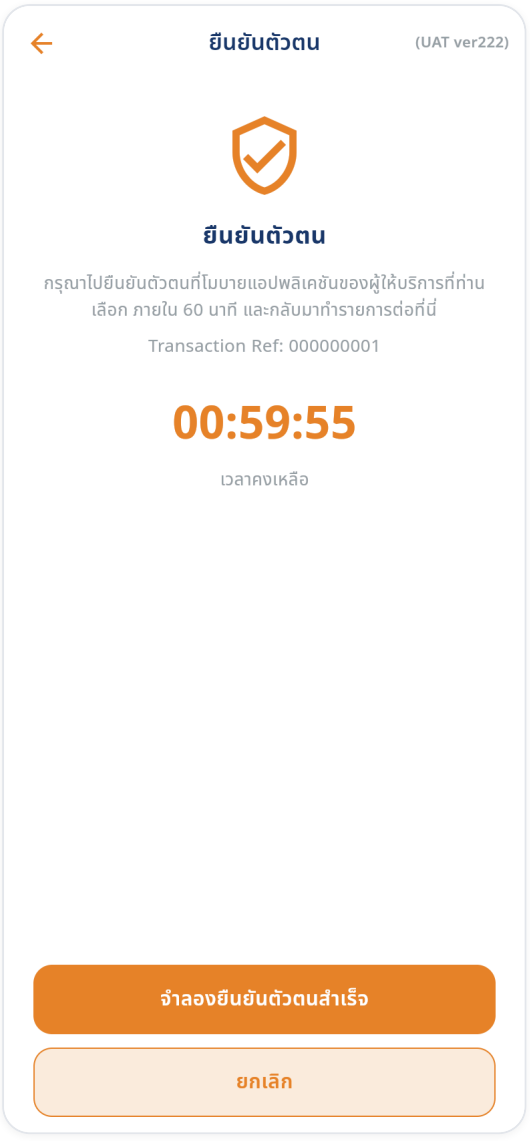
State	What shows
Loading	กำลังโหลดรายชื่อผู้ให้บริการ...
Load failed	ไม่สามารถโหลดรายชื่อผู้ให้บริการ NDID ได้ + **ลองใหม่**
Loaded	Two grids: **ผู้ให้บริการที่เคยลงทะเบียน NDID** and **ผู้ให้บริการที่ยังไม่ลงทะเบียน NDID**

State	What shows
No registered banks	The first grid shows ไม่พบผู้ให้บริการที่ท่านเคยลงทะเบียน NDID; the second grid can still be picked from

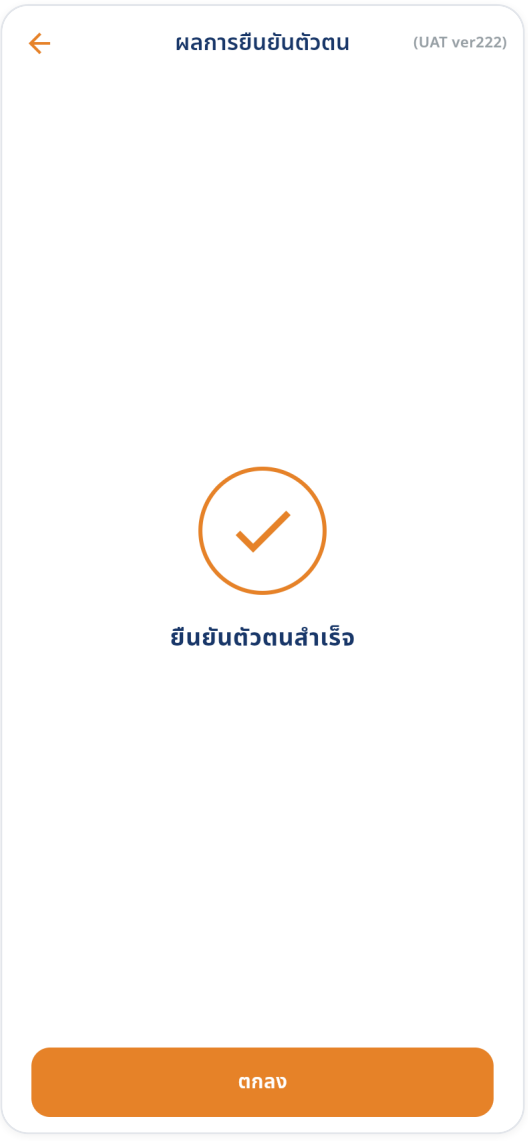
- **Registered** = `/idp/list` called **with** the customer's Thai ID. **Not registered** = the full list minus those.
- A note under the second grid says an unregistered bank must be registered in that bank's app first. The verify request is the same for both grids.
- **Tiles:** the gateway's logo appears on at most **4** registered tiles. Every other tile shows initials (e.g. `MA1`). Each logo is a platform view, and 16 of them crashed the iOS WebView.
- **ถัดไป** is enabled once a bank is picked. It stores `ndidIdpId` and `ndidIdpName`, then pushes Screen 3.
- **ย้อนกลับ** goes back to Screen 1.

Screen 3 — Verify

The app creates a verification request with a 1-hour timeout, then checks its status until the bank app answers. One page has two phases: **waiting** and **success**.



Screen 3 — waiting (plain browser)



Screen 3 — success

These are browser captures. Inside the app, the waiting screen shows the gateway's Transaction Ref instead of `000000001`, and ตรวจสอบสถานะ replaces the จำลองยืนยันตัวตนสำเร็จ button.

State	What shows	Buttons
Creating	กำลังสร้างรายการยืนยันตัวตน...	ยกเลิก

State	What shows	Buttons
Waiting	Standard message [3]: *กรุณาไปยืนยันตัวตนที่โมบายแอปพลิเคชันของผู้ให้บริการที่ท่านเลือก ภายใน 60 นาที และกลับมาทำรายการต่อที่นี่*, plus **Transaction Ref** (from the gateway) and a 60:00 countdown labelled เวลาคงเหลือ	ตรวจสอบสถานะ, ยกเลิก (+ ดู Request Body (debug), non-prod only)
Status checks failing	Orange warning after 3 failures in a row; on HTTP 429: ระบบจำกัดจำนวนการตรวจสอบ…	same as Waiting
Failed	Red NDID Common Message for the IdP/AS `error_code` (see below)	ลองใหม่, ยกเลิก
Timed out (00:00)	หมดเวลาการยืนยันตัวตน กรุณาทำรายการใหม่; the status checks stop	ลองใหม่, ยกเลิก
Success (`ACCEPTED`)	Check icon + **ยืนยันตัวตนสำเร็จ**; `ndidReferenceId` is saved	ตกลง → returns `true`

Status check timing

- Every **3 s** for the first 30 s, then every **15 s**. After an HTTP 429, the next check waits **60 s**. This keeps the app under the gateway limit of 100 requests per 15 minutes.
- The app also checks **immediately** when the customer returns from the bank app. A WebView in the background pauses its timers, so this is needed.
- **ยกเลิก** tries to close the request (if that fails, nothing else happens) and returns `false`.

API calls

Inside the app, every NDID call goes through the host's `httpRequest` bridge. The gateway sends no CORS headers, so a plain browser cannot make these calls.

Screen	Call	Purpose
2	`POST /idp/list` ×2	Registered banks (with the Thai ID) and all banks; filter IAL 2.3 / AAL 2.2
3	`POST /idp/list` + `GET /services/001.cust_info_001/as`	Find the data source (AS) at the same bank as the chosen IdP
3	`POST /rp/verify-with-data`	Create the request (mode 2, 1 h timeout) and ask for customer info. Falls back to `POST /rp/verify` when no AS matches

Screen	Call	Purpose
3	`GET /rp/verify/{reference_id}`	Check the status: `PENDING` / `ACCEPTED` / `REJECTED` / `TIMEOUT` / `CANCELLED` / `REQUESTED_ERROR` / `IDP_OR_AS_ERROR`
3	close request	Best-effort, on ยกเลิก

- **Base URL:** `api_url.ndid_url_base` in Firestore. uat uses `https://uat.ndid.srisawadpower.com`, prod uses `https://ndid.srisawadpower.com`.
- **Auth:** an `X-API-Key` header, chosen to match the gateway's host.
- **Callback:** the AS data goes to `<gateway>/ndid/callback`, which is the backend. The app never reads it back.
- **What the app does with the result:** only `reference_id` goes on to `POST /ploan`, as `ndid_reference_id`.

Error messages

Every failure message is quoted from the NDID standard (005, §6.2.1, pp. 31–39). The app does not write its own wording. Mapping is done in `NdidCommonMessage` (`lib/services/ndid_common_message.dart`).

Input	Message
IdP `error_code` 30000–30900 (13 codes)	That code's standard row
AS `error_code` 40000–40500 (5 codes)	That code's standard row; "RP Contact" becomes ติดต่อผู้ให้บริการ until `NDID_RP_CONTACT` is set
`CANCELLED`	[5] ท่านได้ยกเลิกรายการคำขอหรือเปลี่ยนผู้ให้บริการยืนยันตัวตน กรุณาเลือกผู้ให้บริการยืนยันตัวตนรายอื่น
`REJECTED` / `TIMEOUT` with no code, or an unknown code	p.39 general message: ท่านไม่สามารถทำรายการได้ในขณะนี้ กรุณาทำรายการใหม่หรือเลือกผู้ให้บริการยืนยันตัวตนรายอื่น

The app never shows the word "IdP" or a node ID to the customer. It shows the bank's marketing name instead (§6.2.1 bullet 4).

Environment differences

The real API is used only inside the srisawad app (`NativeCameraBridge.isSupported`). A plain browser runs a simulated version.

	Inside the app	Plain browser
Bank grid	Live list from `/idp/list`	Hardcoded mock banks (BBL, KTB, SCB, …)
Verify	Real request + status checks	A **จำลองยืนยันตัวตนสำเร็จ** button; Transaction Ref `000000001`
`ndidReferenceId`	Set on `ACCEPTED`	Never set

	uat	prod
Gateway	`uat.ndid.srisawadpower.com`	`ndid.srisawadpower.com`
Data source (AS)	Pinned with `ndid_as_id_uat`	Looked up from the chosen bank
Debug button (ดู Request Body)	Shown	Hidden

Known gaps

- ☐ **Browser bypass:** จำลองยืนยันตัวตนสำเร็จ marks the customer as verified with no NDID traffic at all, in any browser that opens the deployed URL. Fix: an `NDID_SIMULATE` define that defaults to off.
- ☐ **Server-side check (pentest #11):** `/ploan` must confirm `ndid_reference_id` with NDID and tie it to the caller's citizen ID. This is the API team's work.
- ☐ **Decline log:** ปฏิเสธ is logged only for the session. A durable record needs an endpoint.
- ☐ **`NDID_RP_CONTACT` :** set a real call-centre number for the 4 AS messages.
- ☐ **Error-code injection:** IdP codes 30000/30200/30400/30700/30900 and all AS codes cannot be triggered from a handset; ask NDID for help.
- ☐ **P-Loan is parked**, so NDID is not in production today.